

BST Physics Curriculum Vol 6 Chapter 4 — Free Energies + Maxwell Relations v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

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Vol 6 Chapter 4 — Free Energies + Maxwell Relations

Chapter motivation

Standard free energies: Helmholtz $F = U - TS$, Gibbs $G = H - TS$, enthalpy $H = U + PV$. Maxwell relations from second-derivative cross-equality. BST identifies free energies as committed-vs-residual substrate-mode balance + Maxwell relations from substrate-symmetry constraints.

Section 4.0b — Reader-grade 3-level pedagogy

Level 1: Free energies $F = U - TS$ = substrate committed-vs-residual mode balance per T2106 eigentone framework; Maxwell relations from substrate-symmetry constraints (Onsager-like reciprocity).

Level 2 (graduate): $F = U - TS$: U = total internal energy = substrate-tick Casimir spectrum integration; TS = “disorder cost” = temperature $\times$ Shannon entropy = substrate-tick computational rate $\times$ per-tick state count. F = useful energy = committed substrate-modes (Vol 4 Ch 2 T2106 eigentone framework). Gibbs $G = H - TS$ with $H = U + PV$ including pressure-volume work; substrate-cartography: pressure = substrate-tick boundary-force rate; volume = substrate Zone-2 coverage extent. Maxwell relations: $(\partial F / \partial V)_T = -P$; $(\partial F / \partial T)_V = -S$; second-derivative cross-equality $(\partial^2 F / \partial T \partial V) = (\partial^2 F / \partial V \partial T)$ gives Maxwell relation $(\partial S / \partial V)_T = (\partial P / \partial T)_V$ — substrate-symmetry constraint preserved.

Level 3 (5th-grader): Free energy $F = U - TS$ = “useful energy” = total energy minus disorder cost. BST: total energy = committed substrate modes; disorder cost = uncommitted modes (residual). Maxwell relations (cross-derivative equalities)

come from substrate symmetry — substrate doesn't care if you change T then V or V then T .

Section 4.1-4.5

Standard free energy definitions; substrate committed-vs-residual reading; Maxwell relations as substrate-symmetry constraints; cross-link Vol 4 Ch 2 eigentone framework.

— Lyra, Vol 6 Ch 4 v0.3, Saturday 2026-05-23 EDT